Government of Karnataka

GOVERNMENT POLYTECHNIC
DEPARTMENT OF TECHNICAL EDUCATION KARNATAKA

Department of Collegiate & Technical Education

SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC

BENGALURU - 560001

2025-26

A Report on

"INTERNSHIP"

Submitted in partial fulfilment for the Award of Diploma in

COMPUTER SCIENCE AND ENGINEERING

by

Medha U Gowda

(102CS23030)

Under the Supervision of

Mr. Abhishek Kumar Anshu

K-AKA Technology Services, Bengaluru

Under the Guidance of

Mr. Raghavendra M Y

Government of Karnataka

GOVERNMENT POLYTECHNIC
DEPARTMENT OF TECHNICAL EDUCATION KARNATAKA

Department of Collegiate & Technical Education

SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC

BENGALURU - 560001

2025-26

CERTIFICATE

This is to certify that **Report** submitted by **Medha U Gowda**, (**102CS23030**) represents his work during the **2025-26** academic year at **K-AKA Technology Services, Bengaluru**, under the Specialization Pathway **"INTERNSHIP(AI&ML)-20CS61S"** fulfilling the requirements for the award of the Diploma in Computer Science and Engineering.

Cohort Owner	Program Coordinator	Principal
Mr. Raghavendra M Y	Mrs. Haritha S M	Sri Harish M

Examiners:

1._____ 2. _____

DATE-10/04/2026

K-AKA TECHNOLOGY SERVICES

1st Cross Road, 6, Outer Ring Rd, Banaswadi, Bengaluru, Karnataka 560043

Intern ID: KTS7937

info@k-akatechnology.in

Ref ID: KTS/INTERNSHIP/2024-2025/KTS7937

www. k-akatechnology.in

Certificate NO: KTS/CERTIFICATE/2024-2025/93104

CERTIFICATE OF INTERNSHIP

This is to certify that Ms. Medha U Gowda (102CS23030), a student of "CSE" from "Sri Jayachamarajendra Government Polytechnic" has completed internship in Software Department (Using AIML) at K-AKA Technology Services (KTS) from 18th December 2025 to 11th April 2026 (16 Weeks) under the guidance of Mr. Abhinav, Software Engineer. During the period of her internship program with us, she was found punctual and creative.

MANAGER
K-AKA Technology Services
Bengaluru, Karnataka

K-AKA TECHNOLOGY SERVICES
Software Company
BENGALURU

DIRECTOR
ABHISHEK ANSHU
K-AKA Technology Services
Bengaluru, Karnataka

MSME
सूक्ष्म, लघु एवं मध्यम उद्यम
MICRO, SMALL & MEDIUM ENTERPRISES

INTERNSHIP REPORT

K-AKA TECHNOLOGY SERVICES

ACKNOWLEDGEMENT

I extend my deepest gratitude to **Sri Jayachamrajendra Government Polytechnic** College, Department of Computer Science and Engineering, for granting me the invaluable opportunity to complete this internship as part of my 6th Semester Diploma curriculum.

I am profoundly thankful to **Sri Harish M**, Principal, for his unwavering encouragement and for fostering a conducive academic environment.

I express my sincere appreciation to **Mrs. Haritha S M**, Head of the Department, for her insightful guidance and steadfast support throughout this internship.

I am particularly indebted to **Mr. Raghavendra M Y**, Cohort Owner, for his expert guidance, motivational support, and constructive feedback that were instrumental in the successful execution of this internship.

I also wish to acknowledge **Mr. Kishore Kumar**, Placement Officer, for his pivotal role in securing this internship and facilitating essential industry exposure.

My heartfelt thanks go to K-AKA Technology Services and its entire team for graciously providing me with this prestigious internship opportunity. I am especially grateful to **Mr. Abhishek Kumar Anshu**, Head of the Organization, for his visionary leadership and profound insights. I further extend my sincere thanks to **Mr. Abhinav Kumar Anshu**, Software Engineer and Technical Guide, for his dedicated technical guidance and unwavering support during my tenure.

I would like to acknowledge the support of my team members and friends for their cooperation and encouragement during the project. Finally, I express my heartfelt gratitude to my parents for their constant support, motivation, and encouragement, which helped me successfully complete this internship.

MEDHA U GOWDA
REG NO: 102CS23030

EXECUTIVE SUMMARY

This report presents the work carried out during the internship at **K-AKA Technology Services**, a Bengaluru-based software development company specializing in SaaS solutions, web applications, and technical training programs.

The project addresses the critical challenge of early identification of genetic disease risk using patient data. Traditional methods of genetic analysis are often time-consuming, costly, and heavily dependent on expert interpretation, making rapid and accessible risk assessment difficult in routine clinical settings. There exists a clear need for an automated, intelligent system capable of analyzing complex genetic information and delivering timely, interpretable risk assessments.

To overcome this, a machine learning-based system was developed to predict an individual's risk of developing a genetic disease, classifying outcomes into three risk categories: **High, Medium, and Low**. The system utilizes **XGBoost (Extreme Gradient Boosting)** as the primary machine learning algorithm, chosen for its resistance to overfitting, and ability to handle feature interactions inherent in genetic datasets.

The web-based application was developed using **Python, Flask, and Tkinter**, integrating three core functional modules: a **Dataset Upload** module for ingesting patient records, a **Prediction Module** for generating risk assessments, and a **Data Visualization Module** featuring charts and graphs.

The project operates within the **healthcare domain**, with the goal of providing a user-friendly platform that supports early clinical decision-making by making genetic risk analysis more accessible. The final system successfully delivers accurate, tiered risk predictions alongside visual outputs, significantly enhancing usability and accessibility. It demonstrates the practical application of machine learning in preventive healthcare and lays a strong foundation for future advancements in predictive genomic systems, including integration with real clinical datasets and expansion to disease-specific risk models.

TABLE OF CONTENTS

	CHAPTERS	Page No
	Chapter-1 Company Profile	**1-8**
1.1	Overview of the Organisation	1-2
1.2	Vision and Mission	2
1.3	Organizational Structure	2-3
1.4	Roles and Responsibilities	4
1.5	Development Process	4-5
1.6	Products and Services	6
1.7	Market Performance and Business Growth	6-7
1.8	Nature of Business	7
1.9	Certifications and Registration	7-8
1.10	Conclusion	8
	Chapter-2 On Job Training 01	**9-12**
2.1	Overview of Job Role	9
2.2	Application of skills and technical knowledge	9
2.3	Tasks Performed and Implementation	10
2.4	Challenges faced and Solutions	10-11
2.5	Performance and Learning Outcomes	11
2.6	Value Addition to Organization	11
2.7	Tools and Technologies Justification	12
2.8	Job Role Understanding and Relevance	12
2.9	Conclusion	12
	Chapter-3 On Job Training 02	**13-16**
3.1	Overview of Job Role	13
3.2	Application of skills and technical knowledge	13
3.3	Tasks Performed and Implementation	14
3.4	System Enhancements and Improvements	14-15
3.5	System Architecture Overview	15
3.6	Performance and Learning Outcomes	15-16
3.7	Value Addition to Organization	16
3.8	Job Role Understanding and Relevance	16
3.9	Conclusion	16
	Chapter-4 Use Cases	**17-29**
4.1	**Use Case-01**	**17-21**
4.1.1	Use Case Title	17
4.1.2	Description	17
4.1.3	Actors	17
4.1.4	Inputs	17
4.1.5	Process	17-19
4.1.6	Output	19
4.1.7	System Workflow	20
4.1.8	Advantages	20

4.1.9	Limitations	20
4.1.10	Future Enhancements	21
4.1.11	Conclusion	21
4.2	**Use Case-02**	**22-29**
4.2.1	Use Case Title	22
4.2.2	Description	22
4.2.3	Actors	22
4.2.4	Inputs	22
4.2.5	Process	23-27
4.2.6	Output	27-28
4.2.7	System Workflow	28
4.2.8	Advantages	28-29
4.2.9	Limitations	29
4.2.10	Future Enhancements	29
4.2.11	Conclusion	29
	Testing	**31-32**
5.1	Test Case 01	31
5.2	Test Case 02	31
5.3	Test Case 03	31
5.4	Test Case 04	32
5.5	Test Case 05	32
5.6	Testing Summary	32
	Model Comparison and Selection	**33**
6.1	Model Accuracy Comparison	33
6.2	Discussion	33
	Results and Discussions	**35-44**
7.1	System Interface Results	33-42
7.2	Model Performance Evaluation	43-44
	Resume	**45**
	Photo Gallery	**47**
	References	**49**
	Appendices	**51**

LIST OF FIGURES

Figure No	Figure Name	Page No
1.1	Company logo	1
1.2	Organization Structure	3
1.3	Development Workflow	5
1.4	Atomm	6
1.5	Skilledhire	6
1.6	Give	6
4.1.1	Backend Initialization	18
4.1.2	Static Output	19
4.1.3	System Workflow	20
4.2.1	CAPTCHA Validation	23
4.2.2	Gauge Meter with Accuracy Bar Chart	28
4.2.3	Final System Workflow	28
7.1.1	Signup Page	33
7.1.2	Login Page	33
7.1.3	Dashboard Page	34
7.1.4	Personal Information and Delete Account pop up	34
7.1.5	Update Page Module	35
7.1.6.1	Before dataset upload	35
7.1.6.2	After dataset upload	35
7.1.7	Dataset Information Page	36
7.1.8	Input fields	36
7.1.9	Final Output	37
7.1.10	History Page	37
7.1.11	Navigation	37
7.1.12.1	Genetic Information Page	38
7.1.12.2	External Integration	38
7.1.13.1	Model Comparison	39
7.1.13.2	Classification Report	39
7.1.14	Backend Output	39
7.1.15	PDF Report	40
9.1	Photo Gallery	47

LIST OF TABLES

Table No	Table Name	Page No
5.1	Test Case 1	30
5.2	Test Case 2	30
5.3	Test Case 3	30
5.4	Test Case 4	31
5.5	Test Case 5	31
6.1	Model Comparison	32
7.2.1	Confusion Matrix	41
7.2.2	Classification Report	41
11.1	Sample Dataset	46

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CAPTCHA	Completely Automated Public Turing test to tell Computers and Humans Apart
CEO	Chief Executive Officer
CGPA	Cumulative Grade Point Average
CTO	Chief Technology Officer
CV	Cross-Validation
DBMS	Database Management System
GUI	Graphical User Interface
HTML	HyperText Markup Language
iPC	Innovative Project Center
ISO	International Organization for Standardization
KTS LMS	K-AKA Technology Services Learning Management System
ML	Machine Learning
NCBI	National Center for Biotechnology Information
NLP	Natural Language Processing
OJT	On-the-Job Training
OTP	One-Time Password
PDF	Portable Document Format
QA	Quality Assurance
REST	Representational State Transfer
SaaS	Software as a Service
SNP	Single Nucleotide Polymorphism
SVM	Support Vector Machine
UI	User Interface
XGBoost	eXtreme Gradient Boosting

CHAPTER 1

COMPANY PROFILE

1.1 Overview of the Organization

Figure 1.1: K-AKA Company Logo

K-AKA Technology Services is a Bengaluru-based software development company specializing in providing end-to-end digital solutions. Established in 2020, the company operates as a Software as a Service (SaaS) provider, focusing on delivering scalable, secure, and high-performance web and mobile applications.

The organization offers a wide range of services including custom website development, enterprise application solutions, mobile application development, and long-term system maintenance. It caters to start-ups, small and medium enterprises, and growing businesses by delivering tailored software solutions that meet specific client requirements.

In addition to its service-oriented business model, the company plays a significant role in technical education by offering structured internship programs. These programs provide students and young professionals with hands-on experience in real-time projects, bridging the gap between academic knowledge and industry expectations.

With a strong emphasis on innovation, usability, and performance, K-AKA Technology Services aims to support digital transformation while simultaneously fostering skill development in the technology sector.

1.2 Vision and Mission

1.2.1 Vision

"Our vision is to become the most trusted IT partner, creating smarter and more connected digital solutions by continuously innovating and advancing technology."

The organization envisions building a strong reputation as a reliable technology partner by delivering scalable solutions while contributing to the growth of skilled professionals in the industry.

1.2.2 Mission

The mission of K-AKA Technology Services is to deliver customized, high-quality software solutions that effectively address client challenges.

Key mission objectives include:

- Providing modern SaaS-based solutions that support business growth
- Delivering efficient and scalable software systems
- Bridging the gap between complex business problems and user-friendly applications
- Supporting learners and professionals through practical technical training

1.3 Organizational Structure

K-AKA Technology Services follows a structured organizational framework that ensures efficient coordination between technical and operational teams.

1.3.1 Executive Leadership

- **Founder / CEO:** Responsible for strategic planning, decision-making, and overall business direction

1.3.2 Technical Leadership

- **CTO / Technical Lead:** Manages system architecture, technical strategy, and development processes

1.3.3 Operational and Support Units

- **Program Managers:** Oversee project planning, project teams and execution
- **Internship Coordinators:** Manage interns and training programs
- **Accounts & Administration:** Handle financial and administrative operations

1.3.4 Technical Wing

The technical department consists of:

- Full Stack Developers
- Mobile Application Developers
- Software Engineers
- QA & Support Engineers

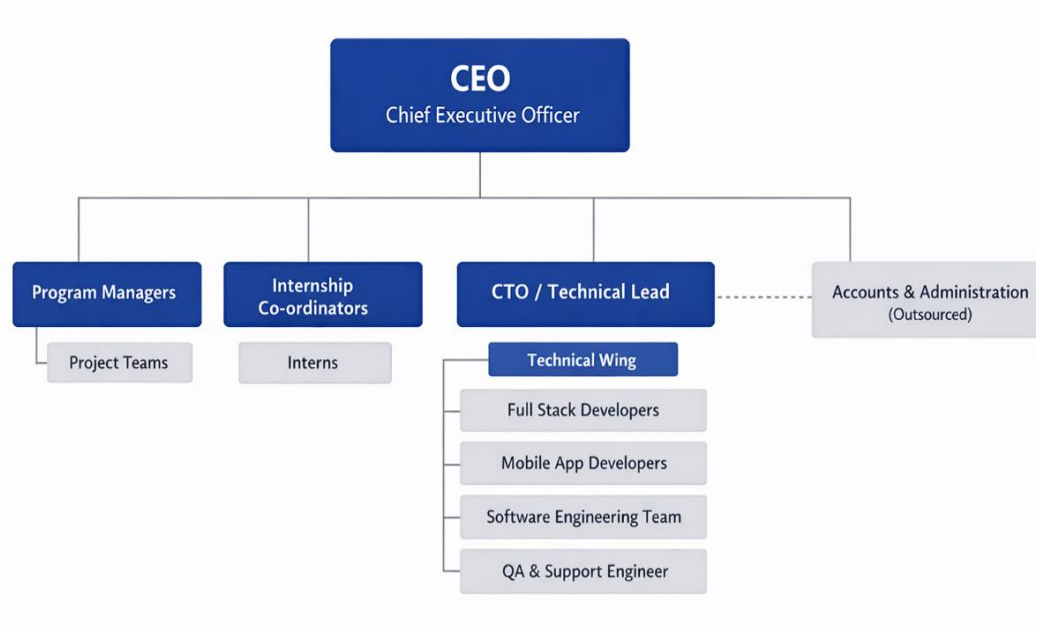


Figure 1.2: Organization Structure

This hierarchical structure ensures smooth workflow, efficient communication, and effective project delivery.

1.4 Roles and Responsibilities

The organization assigns clear roles and responsibilities across different levels:

1.4.1 Leadership Roles

- **CEO:** Defines vision, business strategy, and organizational goals
- **CTO:** Designs system architecture and ensures technical quality

1.4.2 Technical Team

- **Software Engineers:** Develop and maintain software systems
- **Full Stack Developers:** Handle both frontend and backend development
- **Mobile Developers:** Build and optimize mobile applications
- **QA Engineers:** Perform testing, debugging, and system validation

1.4.3 Operational Team

- **Program Managers:** Coordinate projects and manage timelines
- **Business Development Team:** Handle client communication and project acquisition
- **Accounts Team:** Manage financial operations

1.4.4 Internship and Training Division

- **Interns:** Assist in development tasks and gain practical experience
- **Internship Coordinators:** Guide and monitor intern performance

This division of responsibilities ensures efficient execution of projects and skill development within the organization.

1.5 Development Process

K-AKA Technology Services follows a structured Software Development Life Cycle (SDLC) aligned with SaaS delivery models.

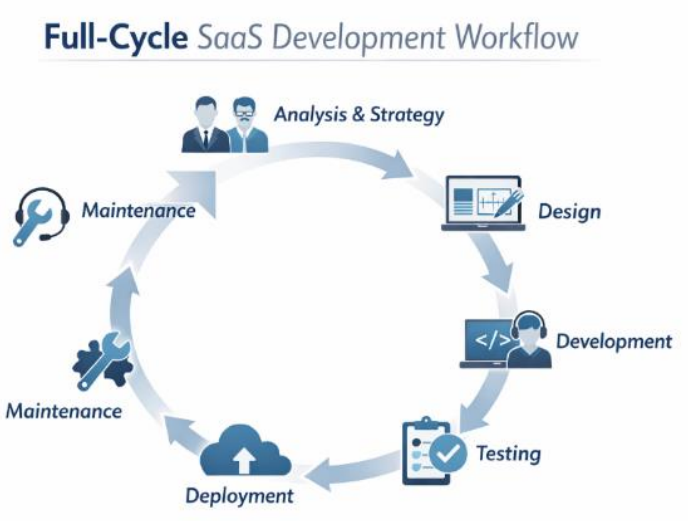


Figure 1.3: Development Workflow

1. Analysis and Planning

Project requirements are gathered and analyzed. Strategic decisions are made regarding system architecture and technology selection.

2. Design

User interface designs and system architecture are developed to ensure usability and scalability.

3. Development

Developers implement the system by writing code, integrating APIs, and building frontend and backend components.

4. Testing

Quality assurance processes are conducted to identify and fix errors, ensuring system reliability and performance.

5. Deployment

The application is deployed on cloud-based platforms and delivered to clients as a SaaS solution.

6. Maintenance and Support

Continuous monitoring, updates, and performance improvements are carried out to maintain system efficiency.

This structured approach ensures high-quality product delivery and long-term system sustainability.

1.6 Products and Services

Client Portfolio

K-AKA lists clients including – Atomm, Skilledhire, Give

Figure 1.4: Atomm **Figure 1.5: Skilledhire** **Figure 1.6: Give**

1.6.1 Core Services

- Custom website development
- Web and mobile application development
- Enterprise software solutions
- System maintenance and technical support
- Technical training and internship programs

1.6.2 Products Offered

- **MyBusiness:** Business management software
- **KTS LMS:** Learning Management System for training and education
- **iPC (Innovative Project Center):** Platform for project-based learning

These offerings highlight the company's focus on both software development and technical education.

1.7 Market Performance and Business Growth

K-AKA Technology Services has demonstrated steady growth since its establishment.

1.7.1 Key Highlights

- Successful delivery of multiple web and application projects

- Strong client retention through continuous support services
- Expansion from basic web development to full-cycle software solutions

1.7.2 Market Presence

- Serves startups, small businesses, and enterprises
- Operates at local, national, and remote levels
- Builds long-term client relationships

1.7.3 Competitive Advantages

- Agile development approach
- Use of modern technologies
- Personalized client engagement

1.7.4 Future Growth

- Expansion into advanced SaaS solutions
- Increasing demand for maintenance services
- Growth through referrals and digital presence

1.8 Nature of Business

K-AKA Technology Services operates as a software services and training organization.

Its core business activities include:

- Development of cloud-based software applications
- Delivery of SaaS-based solutions
- Providing technical internships and training programs
- Supporting clients through development and maintenance services

The company follows a service-oriented model that combines technology solutions with skill development initiatives.

1.9 Certifications and Registration

K-AKA Technology Services is recognized for maintaining quality standards and operational credibility. The company is certified under **ISO**

9001:2015, which reflects its commitment to quality management systems and continuous improvement in service delivery.

Additionally, the organization is officially registered with the Government of India under registration number **KR-03-0010654**, ensuring its compliance with regulatory and legal requirements.

1.10 Conclusion

K-AKA Technology Services is a dynamic and emerging software development organization that effectively combines technology solutions with skill development initiatives. The company has established itself as a reliable provider of scalable and user-centric digital solutions while also contributing to the growth of aspiring professionals through structured internship programs.

With a strong organizational structure, well-defined development processes, and a commitment to quality standards such as ISO 9001:2015 certification, the company continues to enhance its service capabilities and market presence.

Its focus on innovation, customer satisfaction, and continuous improvement positions it well for future growth in the competitive software and SaaS industry. The organization demonstrates a balanced approach toward business development and educational impact, making it a valuable contributor to the technology ecosystem.

CHAPTER 2

ON JOB TRAINING-01

2.1 Overview of Job Role

Job Role: AIML and Software Development Intern

Responsibility: Development

During the initial phase of the internship, the role undertaken was that of an **AI/ML and Software Development Intern**, focusing on the development of a machine learning-based web application for predicting genetic disease risk.

The objective of this phase was to build the foundational components of the system, including user authentication, dataset handling, backend integration, and basic machine learning workflow. The system aimed to assist users in analyzing genetic risk factors using structured patient data and family history.

2.2 Application of Skills and Technical Knowledge

A wide range of technical skills and tools were applied during this phase:

- **Programming Language:** Python
  Used for backend logic, API development, and data processing.
- **Libraries:** Pandas, Scikit-learn
  Pandas was used for dataset cleaning, transformation, and handling missing values.
  Scikit-learn was used for implementing initial machine learning models.
- **Backend Framework:** Flask
  Flask was used to design REST APIs for handling authentication, dataset processing, and prediction requests.
- **Frontend Technologies:** Tkinter, HTML, CSS, JavaScript
  Tkinter was used for initial GUI interfaces, while web technologies were used to design interactive dashboards and prediction pages.
- **Database:** Firebase
  Firebase Authentication and Realtime Database were used for secure storage and retrieval of user data.
- **Testing Tool:** Postman
  Used for validating API endpoints and ensuring proper communication between frontend and backend.
- **Development Tools:** PyCharm, Spyder, Sublime Text

2.3 Tasks Performed and Implementation

The development process followed a structured and modular approach:

- Developed a **user authentication system** including login, signup, and CAPTCHA validation to ensure secure access.
- Integrated **Firebase database** for real-time storage of user credentials and profile information.
- Designed and implemented **dynamic user interfaces** for login, signup, and dashboard modules.
- Established **frontend-backend communication** using HTTP requests and API routing.
- Created a **dataset upload module** to allow users to input data in Excel format.
- Implemented **data pre-processing techniques**, including:
    - Handling missing values
    - Removing duplicates
    - Formatting and encoding data
- Developed a **machine learning prediction interface** for processing user inputs.
- Implemented **basic visualization components**, including charts and a risk indicator.
- Performed **API testing using Postman** and debugged errors related to data formatting and API responses.

2.4 Challenges Faced and Solutions

During the development of the initial system, several challenges were encountered, particularly in handling real-world datasets and integrating different system components.

One of the primary challenges was dealing with inconsistent dataset formats, including missing values and irregular data structures. This issue was addressed by implementing robust data preprocessing techniques such as data cleaning, normalization, and encoding.

Another challenge involved establishing seamless communication between the frontend and backend. Initial API integration resulted in errors due to

improper request handling and data formatting mismatches. These issues were resolved through systematic API testing using Postman and debugging of request-response cycles.

Additionally, designing a user-friendly interface while maintaining functionality required multiple iterations. Improvements were made by refining UI components and simplifying navigation.

2.5 Performance and Learning Outcomes

All assigned tasks were completed within the stipulated timeline. The development process involved continuous debugging and iterative improvements, particularly in handling real-world datasets with inconsistencies.

Key learning outcomes include:

- Understanding the integration of machine learning models into web applications
- Gaining practical experience in backend API development
- Learning effective data pre-processing techniques
- Improving debugging and problem-solving skills
- Developing user-friendly interfaces for non-technical users

2.6 Value Addition to the Organization

The work contributed to the organization by developing a functional prototype for genetic disease prediction.

- A structured **data pre-processing pipeline** was established, enabling efficient handling of real-world datasets.
- The system improved **data accessibility and usability** through a simple interface.
- Initial visualization components enhanced the interpretability of prediction results.

This phase laid the groundwork for building a scalable and user-centric predictive system.

2.7 Tools and Technologies Justification

The selection of tools and technologies played a crucial role in the successful development of the system. Python was chosen due to its extensive support for machine learning and data processing libraries. Flask was selected as a lightweight backend framework that enables efficient API development and easy integration with frontend components.

Firebase was used for its real-time database capabilities and secure authentication features, making it suitable for handling user data. Frontend technologies such as HTML, CSS, and JavaScript were used to create interactive and responsive user interfaces.

Overall, the chosen technology stack ensured scalability, flexibility, and ease of development.

2.8 Job Role Understanding and Relevance

The role required a combination of software development and machine learning skills, including:

- Backend development
- Data pre-processing
- Frontend design
- Model integration

This phase helped bridge the gap between theoretical knowledge and practical implementation, providing hands-on experience in developing real-world applications.

2.9 Conclusion

The first phase of the internship focused on building the foundational components of the system, including user authentication, dataset handling, and initial machine learning integration. This phase provided valuable experience in applying theoretical concepts to real-world problems and helped in understanding system architecture, API integration, and data pre-processing techniques. Overall it gave a strong base for further development and improvement.

CHAPTER 3

ON JOB TRAINING-02

3.1 Overview of Job Role

Job Role: AIML and Software Development Intern

Responsibility: Testing and Optimization

The second phase of the internship focused on enhancing and optimizing the previously developed system. The role continued as an **AI/ML and Software Development Intern**, with emphasis on improving system performance, usability, and model accuracy.

This phase transformed the initial prototype into a more refined and user-friendly application by introducing advanced machine learning models, improved UI design, and additional functional modules.

3.2 Application of Skills and Technical Knowledge

Advanced concepts and tools were applied in this phase:

- **Backend Development (Python + Flask):**
  Enhanced API structure for improved efficiency and modularity.
- **Machine Learning Models:**
  Multiple models were evaluated, including:
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Random Forest
  - Gradient Boosting
  - XGBoost

  The final model selected was **XGBoost** due to its superior performance and ability to handle non-linear relationships.
- **Database Management (Firebase):**
  Improved CRUD operations and secure handling of user data.
- **Frontend Enhancements:**
  Implemented modern UI design with responsive layouts, structured cards, and interactive elements.

- **Data Pre-processing Enhancements:**
  Improved feature selection, normalization, and dataset validation mechanisms.

3.3 Tasks Performed and Implementation

Several enhancements and new features were implemented:

- Improved **user authentication system** with better validation and error handling.
- Redesigned the **dashboard interface** with structured layout and improved navigation.
- Added **profile management features**, including update and delete account functionality with authentication checks.
- Implemented **keyboard shortcuts** for faster navigation and improved user experience.
- Enhanced the **dataset module** by adding:
  - Sample dataset download
  - Dataset format guidance
  - Automatic dataset analysis (rows, columns, size)
- Developed a more advanced **prediction module** with:
  - Structured input fields
  - Toggle-based SNP selection
  - Sample data autofill option
- Added **prediction history tracking**, enabling users to view past predictions.
- Implemented an **educational genetic information module** explaining SNP values and gene expression levels.
- Developed **advanced visualization components**, including a dynamic risk gauge meter.
- Added **PDF report generation** for downloadable prediction results.

3.4 System Enhancements and Improvements

Compared to the initial phase, several enhancements were introduced to improve system performance and usability.

One of the major improvements was the transition from basic machine learning models to advanced algorithms such as XGBoost, which significantly improved prediction accuracy and model performance.

The user interface was redesigned using modern design principles, including card-based layouts, responsive structure, and interactive elements. This improved user experience and made the system more accessible to non-technical users.

Additional modules such as prediction history, genetic data information, dataset guidance, and PDF report generation were implemented to enhance system functionality and usability.

Furthermore, validation mechanisms were strengthened to ensure that only correct and meaningful data is processed by the system. Unnecessary restrictions on name fields, were also refined.

3.5 System Architecture Overview

The system follows a multi-layered architecture consisting of frontend, backend, database, and machine learning components.

The frontend layer handles user interaction and interface rendering, while the backend layer manages business logic and API communication. The database layer, implemented using Firebase, is responsible for storing user data and application records.

The machine learning layer processes input data, performs predictions, and generates outputs. These layers are interconnected through API calls, ensuring smooth data flow and system functionality.

3.6 Performance and Learning Outcomes

This phase demonstrated significant improvement in both technical and design aspects of the system.

Key learning outcomes include:

- Understanding model comparison and selection techniques

- Implementing advanced machine learning algorithms (XGBoost)
- Designing user-centric interfaces with improved usability
- Enhancing system security and validation mechanisms
- Developing complete end-to-end application workflows

3.7 Value Addition to the Organization

This phase contributed significantly by transforming the system into a more practical and scalable solution:

- Improved **model accuracy and reliability** through advanced algorithms
- Enhanced **user experience** with intuitive design and accessibility features
- Reduced dependency on external datasets by introducing **sample dataset functionality**
- Added **prediction history and reporting features**, improving usability and record-keeping

3.8 Job Role Understanding and Relevance

The role in this phase required a deeper understanding of system design and optimization, including:

- Advanced machine learning implementation
- Full-stack development
- User experience enhancement
- System integration and performance optimization

This phase reinforced the importance of combining technical knowledge with practical usability considerations in real-world applications.

3.9 Conclusion

The second phase of the internship emphasized system enhancement, optimization, and user experience improvement. Advanced machine learning models such as XGBoost were implemented to improve prediction accuracy, and additional features like dataset guidance, prediction history, and report generation were introduced. This phase strengthened skills in full-stack development, system design, and performance optimization. Overall, OJT–02 transformed the initial prototype into a comprehensive and user-friendly application.

CHAPTER 4

4.1 USE CASE – 01

4.1.1 Use Case Title

Genetic Disease Risk Prediction System

4.1.2 Description

The Genetic Disease Risk Prediction System is a machine learning-based application designed to assist users in analysing the probability of genetic diseases based on patient data and genetic indicators.

The system enables users to securely log in, upload datasets, and perform predictive analysis using machine learning techniques. It provides both numerical and visual outputs, making the results easy to interpret for users with minimal technical knowledge.

4.1.3 Actors

- User (Doctor / Researcher / Admin)
- System

4.1.4 Inputs

The system requires the following inputs for prediction:

- **Age:** Numerical value representing patient age
- **Gender:** Male or Female
- **Family History:** Indicates hereditary influence (Yes/No)
- **SNP Values (SNP1–SNP10):** Genetic variation indicators
- **Gene Expression Values:** Activity levels of specific genes
- **Dataset Upload:** Excel file (.xlsx format) for training and testing

4.1.5 Process

The system follows a structured workflow from user authentication to prediction output:

Step 1: System Initialization and Login

The system is initiated by running the backend and frontend applications. Upon execution, the login interface is displayed.

```
C:\Users\HOME\OneDrive\Desktop\fullstackmlproject\backend>py api.py
 * Serving Flask app 'api'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

Figure 4.1.1: Backend Initialization

Step 2: User Authentication and Account Creation

New users can create an account by providing required details such as name, user ID, email, phone number, and password.

Input validation ensures:

- Proper email format
- Minimum length for user ID
- Matching passwords
- Valid phone number

Step 3: Dashboard Access

After successful login, the user is redirected to the dashboard.

The dashboard displays:

- User profile information (name, contact details)
- Dataset upload section
- Navigation options for further actions

Users can also update their profile details or manage their account through the dashboard interface.

Step 4: Dataset Upload and Pre-processing

Users can upload a dataset in Excel format [See Appendix A]

Once uploaded:

- The dataset is processed automatically
- Data cleaning and formatting operations are applied

- Dataset properties such as rows and columns and column headings except the target column are displayed

This ensures the data is ready for machine learning operations.

Step 5: Prediction Input

The user proceeds to the prediction module and enters required input values, including:

- Personal details
- Genetic markers (SNP values)
- Gene expression levels

Step 6: Machine Learning Prediction

The processed input data is passed to the machine learning model which is not currently incorporated into the system but will be added in phase 2.

Step 7: Output Visualization

The prediction results are displayed on the interface in an easy-to-understand format, which is now currently static displaying only 45%. This visual representation helps users interpret the results effectively.

4.1.6 Output

The system generates the following outputs (currently static):

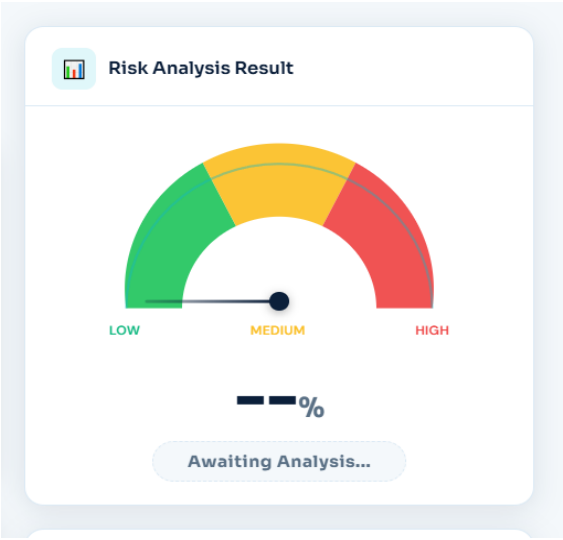


Figure 4.1.2: Static Output

4.1.7 System Workflow

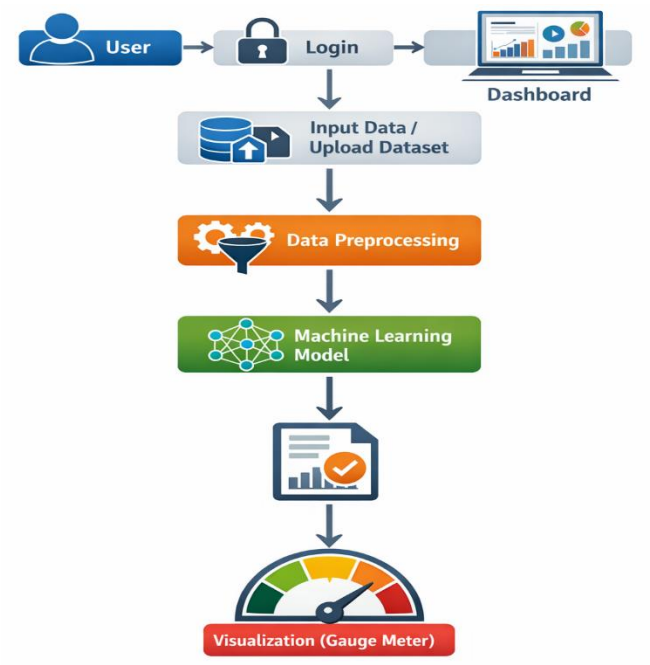


Figure 4.1.3: System Workflow

4.1.8 Advantages

- Enables early identification of potential genetic risks
- Provides user-friendly interface for non-technical users
- Integrates machine learning for accurate prediction
- Supports dataset upload for flexible usage
- Enhances result understanding through visualization

4.1.9 Limitations

- Accuracy depends on dataset quality
- Limited input parameters compared to real-world genetic complexity
- Requires user-provided data for prediction
- Model may not capture all biological variations
- The current system relies on basic authentication mechanisms and does not verify the authenticity of user email addresses.
- The use of a publicly configured Firebase database may introduce potential security risks if not properly managed.

4.1.10 Future Enhancements

- Integration of advanced machine learning models
- Inclusion of additional genetic and environmental parameters
- Improved dataset quality and diversity
- Integration with real-time medical data sources
- Advanced authentication mechanisms such as Google account verification
- One-Time Password (OTP)-based login to improve user authenticity.
- The implementation of a dedicated admin module with role-based access
- Additionally, improvements in database security are required, as the current Firebase configuration may be vulnerable if not properly secured.
- Future work can include implementing stricter database rules and encryption mechanisms to ensure data privacy and protection.

4.1.11 Conclusion

The Genetic Disease Risk Prediction System demonstrates the practical application of machine learning in the healthcare domain.

It successfully integrates user authentication, dataset processing, and predictive analysis into a single platform. While the system has certain limitations, it provides a strong foundation for further development and real-world implementation.

4.2 USE CASE – 02

4.2.1 Use Case Title

Enhanced Genetic Disease Risk Prediction System

4.2.2 Description

The Enhanced Genetic Disease Risk Prediction System is an advanced version of the machine learning-based web application designed to predict genetic disease risk using patient data and genetic indicators.

This version focuses on improving system accuracy, usability, and user experience by integrating advanced machine learning models, enhanced validation mechanisms, and additional functional modules such as prediction history, dataset guidance, and report generation.

The system provides a complete end-to-end workflow, enabling users to securely interact with the application, process datasets, and obtain interpretable prediction results.

4.2.3 Actors

- User (Doctor / Researcher / Admin)
- System

4.2.4 Inputs

The system accepts the following inputs:

- **Age:** Numeric value (validated range)
- **Gender:** Male or Female
- **Family History:** Yes/No indicator
- **SNP Values (SNP1–SNP10):** Toggle-based genetic markers
- **Gene Expression Values (1–5):** Decimal inputs
- **Dataset Upload:** Excel file (.xlsx format)
- **Optional Name Field:** For storing prediction history

4.2.5 Process

The system operates through a comprehensive and user-centric workflow:

Step 1: System Execution

The application is initiated by running both backend and frontend modules. Upon execution, the login interface is displayed, marking the entry point of the system.

Step 2: Authentication Module

The system provides a secure authentication mechanism. The system allowed invalid data to be stored during account creation. This issue was resolved by implementing strict validation checks, ensuring that user data is stored in the database only after successful verification.

- Users log in using user ID, password, and CAPTCHA
- CAPTCHA dynamically refreshes on incorrect attempts
- Error-specific messages are displayed for invalid inputs

Figure 4.2.1: CAPTCHA validation

New users can register through the signup module with strict validation rules:

- Minimum user ID length
- Valid email format
- Password confirmation checks
- Phone number validation

User data is stored in the database **only after successful validation**, ensuring data integrity and preventing invalid entries.

Step 3: Dashboard and User Management

After successful login, the user is redirected to the dashboard.

The dashboard includes:

- User profile information (name, email, contact details)
- Dataset upload interface
- Functional navigation controls

Additional features include:

- **Profile Update:** Users can modify personal details now upgraded with validation with confirmation messages
- **Account Deletion:** Requires password verification before deletion
- **Session Control:** Prevents unauthorized access after account deletion
- **Keyboard Shortcuts:**
    - Ctrl + E → Update profile
    - Ctrl + U → Upload dataset
    - Ctrl + P → Prediction module

These features enhance both usability and system security.

Step 4: Dataset Information and Guidance

The system provides a dedicated dataset information module to assist users in preparing valid datasets.

This module includes:

- Sample dataset download option
- Required column specifications (age, gender, SNP values, gene expression, disease)
- Formatting rules and validation guidelines
- Example dataset structure [See appendix A]

This ensures that users can easily prepare compatible datasets, reducing processing errors.

Step 5: Dataset Upload and Processing

Users can upload datasets through an intuitive interface supporting drag-and-drop functionality. [See Appendix A]

Key operations include:

- File validation (.xlsx format only)
- Real-time upload progress indication
- Automatic dataset processing
- Extraction of dataset properties such as:
    - File size
    - Number of rows and columns
    - Column names

This step prepares the dataset for machine learning operations.

Step 6: Prediction Module

The prediction module is designed with structured and user-friendly input sections:

Personal Information

- Age (validated range)
- Gender selection
- Family history indicator

Genetic Markers

- SNP values (SNP1–SNP10) using toggle buttons

Gene Expression

- Decimal input fields for gene expression values

Additional features:

- **"Fill Sample Data" button** for first-time users
- Input validation to ensure correct data entry

Step 7: Machine Learning Prediction

The system uses an optimized machine learning model (XGBoost) to analyze input data.

- The model processes both user inputs and dataset patterns

- It generates a probability score representing genetic disease risk

This ensures improved accuracy and better handling of complex relationships in data.

Step 8: Output and Visualization

The prediction results are displayed using interactive visualization techniques:

- **Risk Percentage** (e.g., 83.66%)
- **Risk Category:**
  - Low (0–33%)
  - Medium (34–65%)
  - High (66–100%)
- **Gauge Meter Visualization:**
  Displays risk level with color-coded indicators (green, yellow, red)
- **Result Interpretation Section:**
  Explains each risk category for better understanding

Step 9: Prediction History

The system maintains a history of user predictions:

- Stores user-specific prediction records
- Displays:
  - Date and time
  - Input details
  - Risk percentage and category

This allows users to track and review previous analyses.

Step 10: Genetic Information Module

A dedicated module is provided to educate users about genetic parameters:

- Explanation of SNP values and gene expression levels
- Visual representation using activity bars
- Information on factors affecting genetic risk
- External integration with genetic databases (e.g., NCBI)

This improves user understanding and system usability.

Step 11: Model Comparison Module

This module provides users with an overview of different machine learning algorithms evaluated during the development of the system.

- Displays a comparison of multiple machine learning models such as Logistic Regression, SVM, Random Forest, Gradient Boosting, and XGBoost based on accuracy
- Highlights XGBoost as the selected model
- Includes a visual representation using a bar chart
- Performance metrics like accuracy, confusion matrix are added

Step 12: Report Generation

The system allows users to download prediction results as a PDF report.

The report includes:

- Risk percentage and category
- Graphical representation (bar chart)
- Recommendations based on risk level
- Disclaimer regarding medical usage

This feature enhances usability for documentation and sharing purposes.

4.2.6 Output

The system produces:

- Risk Percentage
- Risk Category (Low / Medium / High)
- Gauge Meter Visualization
- Prediction History Records
- Downloadable PDF Report

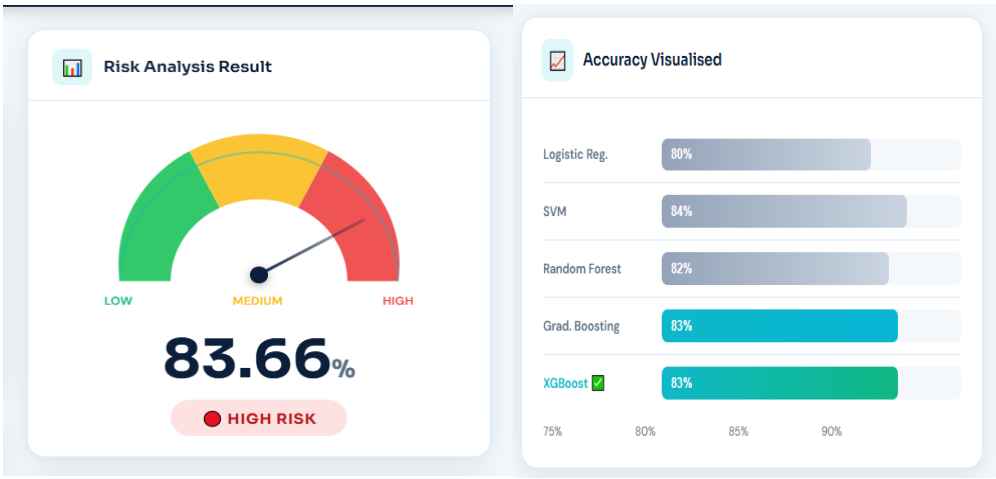


Figure 4.2.2: Gauge Meter with Accuracy Bar Chart

4.2.7 System Workflow

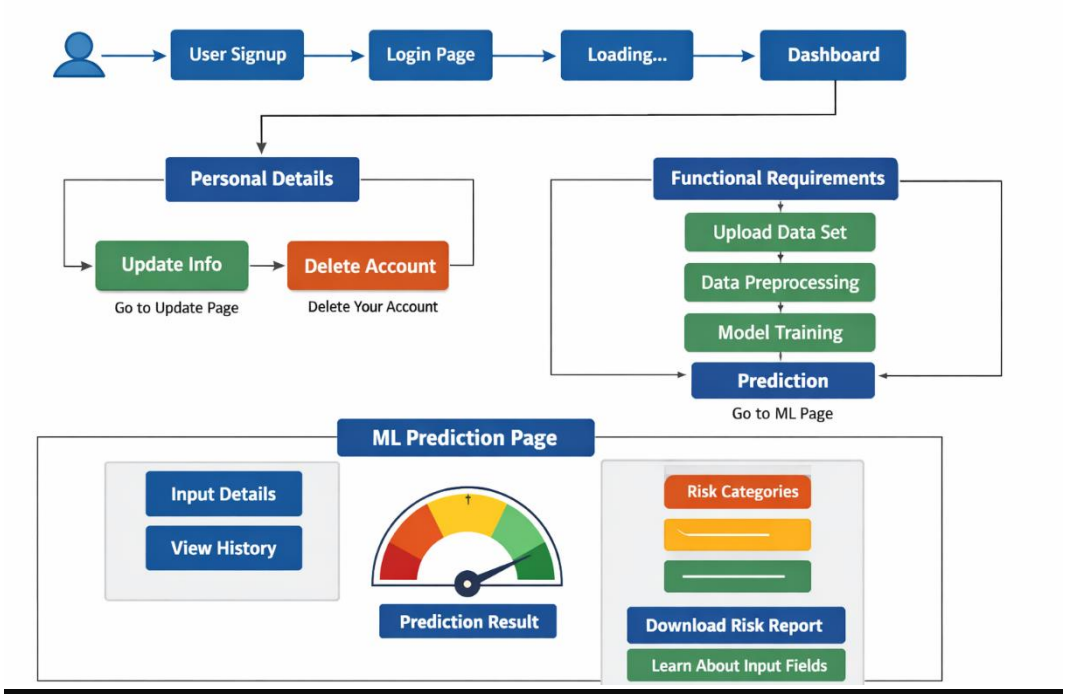


Figure 4.2.3: Final System Workflow

4.2.8 Advantages

- Provides a complete end-to-end predictive system
- Uses advanced machine learning (XGBoost) for improved accuracy
- Offers strong validation and secure authentication

- Enhances user experience through intuitive interface design
- Includes prediction history and report generation
- Provides educational support for genetic data understanding

4.2.9 Limitations

- Model accuracy depends on dataset quality
- Limited representation of complex biological factors
- Relies on user-provided input data
- Not a substitute for professional medical diagnosis
- Full-scale deployment was not implemented due to platform constraints and resource limitations.

4.2.10 Future Enhancements

- Integration of larger real-world genetic datasets
- Implementation of deep learning models
- Mobile application development
- Enhanced security mechanisms
- Real-time healthcare system integration
- Future work includes deploying the system on scalable cloud platforms such as Render or Railway

4.2.11 Conclusion

The Enhanced Genetic Disease Risk Prediction System demonstrates a comprehensive integration of machine learning and web technologies.

The system successfully incorporates advanced features such as dataset guidance, prediction history, visualization, and report generation, making it a robust and user-friendly application.

With further improvements and validation, the system has strong potential for real-world applications in predictive healthcare and personalized medicine.

TESTING

5.1 Test Case 1: Authentication Module Testing

Table 5.1: Test Case 1

Test ID	Scenario	Result	Status
T1	Invalid User ID	"User not found"	**PASS**
T2	Incorrect Password	"Incorrect password"	**PASS**
T3	Invalid CAPTCHA	Error shown and Captcha refreshed	**PASS**
T4	Successful Login	Dashboard access granted	**PASS**
T5	Initial Signup	Data stored even when validation failed	**FAIL**
T6	Updated Signup	Data stored only after successful validation	**PASS**

5.2 Test Case 2: User Management Module Testing

Table 5.2: Test Case 2

Test ID	Scenario	Result	Status
T7	Initial Update profile	No validation, invalid data taken	**FAIL**
T8	Update profile (fixed)	Validation applied for all fields	**PASS**
T9	Data Update	Details saved and reflected	**PASS**
T10	Delete Confirmation	Confirmation prompt shown	**PASS**
T11	Delete Authentication	Requires correct password	**PASS**
T12	Post Deletion Behaviour	User logged out and cannot re-enter	**PASS**

5.3 Test Case 3: Dataset Module Testing

Table 5.3: Test Case 3

Test ID	Scenario	Result	Status
T13	Dataset Upload	Upload works from any location	**PASS**
T14	Progress Indicator	Displayed correctly	**PASS**
T15	Dataset processing	Rows, columns and size shown	**PASS**
T16	Column Display	Feature names shown correctly	**PASS**
T17	Dataset info Page	Opens with correct information	**PASS**
T18	Sample Dataset download	Downloads successfully with .xlsx format	**PASS**

5.4 Test Case 4: Prediction Module Testing

Table 5.4: Test Case 4

Test ID	Scenario	Result	Status
T19	Prediction Page Load	Page loads correctly	**PASS**
T20	Fill Sample Data	Fields auto filled successfully	**PASS**
T21	Input Validation	Inputs validated properly	**PASS**
T22	SNP toggle input	Toggle selection working correctly	**PASS**
T23	Prediction Execution	Prediction generated successfully	**PASS**
T24	Output Display	Risk% and categories displayed	**PASS**
T25	Gauge Meter	Needle moves based on result (High, Medium or Low)	**PASS**

5.5 Test Case 5: Output, History and Reports Check

Table 5.5: Test Case 5

Test ID	Scenario	Result	Status
T26	Prediction History	Displays past predictions correctly	**PASS**
T27	Missing Name handle	Displays unknown when name not provided	**PASS**
T28	PDF Download	Report downloads successfully	**PASS**
T29	PDF Content	Includes risk%, chart ,recommendations and disclaimer	**PASS**
T30	Genetic Info Page	Opens and displays correctly	**PASS**
T31	External Link	Redirects successfully	**PASS**
T32	Back Button Issue	Back button of genetic info not working	**FAIL**

5.6 Testing Summary

The system was tested across all major modules, including authentication, dataset handling, prediction processing, and report generation.

Most functionalities performed as expected, with successful validation, accurate prediction outputs, and smooth navigation across modules. The model's accuracy was confirmed with accuracy score and classification report (83.4% avg).

Minor limitations, such as the non-functional back button in the genetic information module, were observed but do not significantly impact overall system performance. Overall, the system demonstrates reliable performance, robust validation mechanisms, and successful integration of all modules, resulting in a stable and functional application.

MODEL COMPARISON AND SELECTION

To identify the most suitable machine learning algorithm for the genetic disease prediction system, multiple models were evaluated and compared based on their performance. The models considered include Logistic Regression, Support Vector Machine (SVM), Random Forest, Gradient Boosting, and XGBoost.

6.1 Model Accuracy Comparison

Table 6.1: Model Comparison

Model	Accuracy	Reason
Logistic Regression	80%	Limited ability to handle non-linear relationships
SVM	84%	Inconsistent probability Outputs
Random Forest	82%	Moderate performance for given dataset
Gradient Boosting	83%	Less efficient compared to XGBoost
XG Boost	83%	Best Balance of accuracy and efficiency

Although SVM showed slightly higher accuracy, XGBoost was selected as the final model due to its superior balance between performance, speed, and ability to handle complex data relationships. It also provides better scalability and robustness for structured datasets.

A graphical comparison of model accuracies was implemented using a bar chart, allowing users to visually interpret the performance differences between models

6.2 Discussion

The comparison highlights that no single model is universally optimal; however, XGBoost demonstrated consistent performance and reliability. The inclusion of a dedicated model comparison interface enhances system transparency and allows users to understand the reasoning behind model selection.

RESULTS AND DISCUSSIONS

7.1 System Interface Results

7.1.1 Signup Page

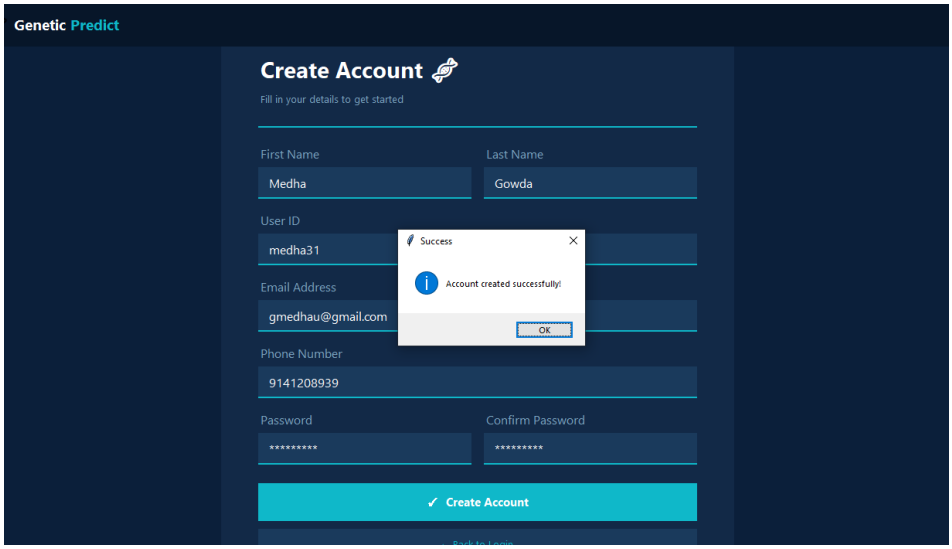


Figure 7.1.1: Signup Page

7.1.2 Login Page

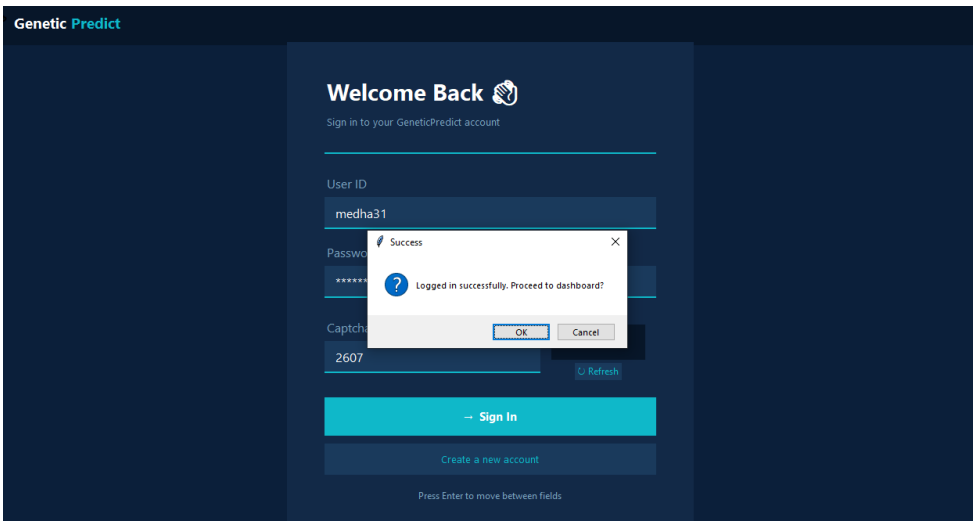


Figure 7.1.2: Login Page

7.1.3 Dashboard

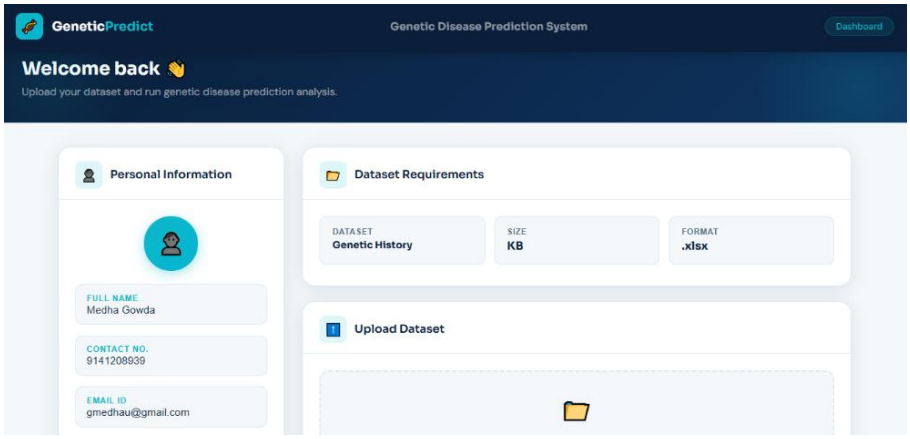


Figure 7.1.3 Dashboard Page

7.1.4 Profile Section

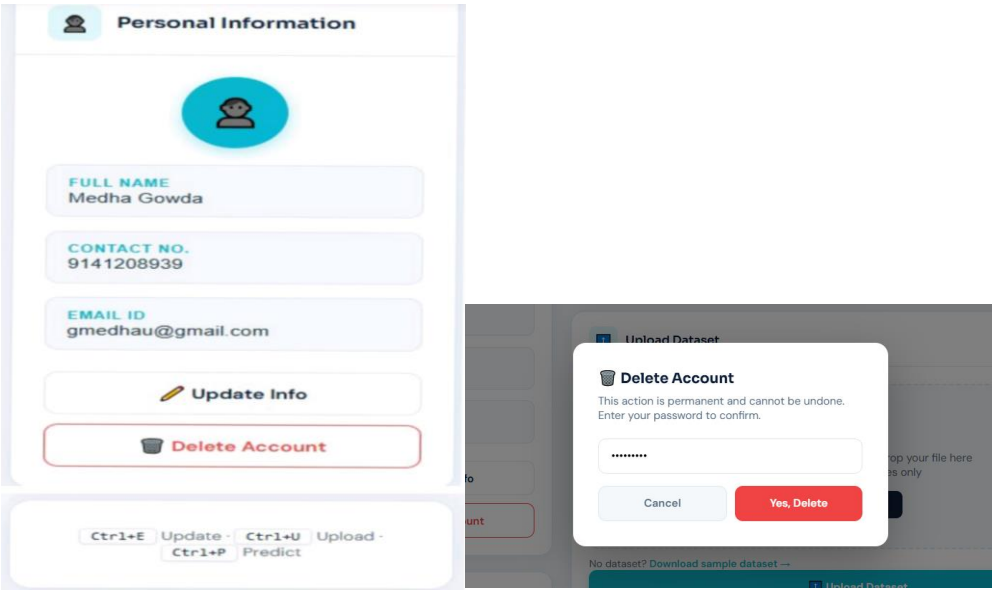


Figure 7.1.4: Personal information and Delete Account pop up

7.1.5 Update Page

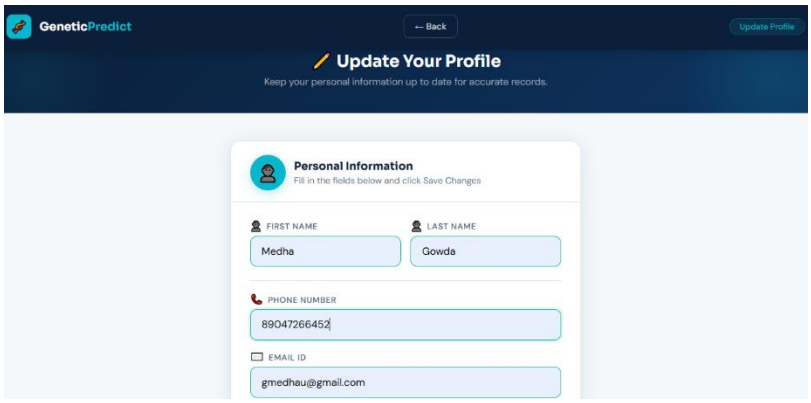


Figure 7.1.5: Update Page Module

7.1.6 Dataset Upload

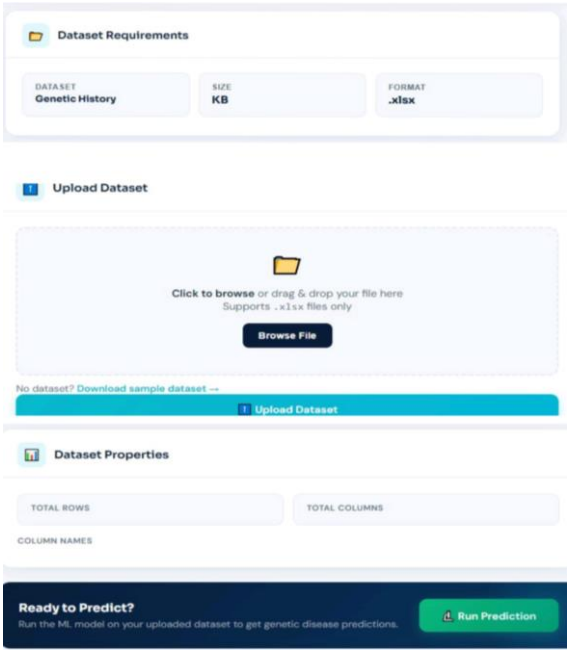


Figure 7.1.6.1 : Before dataset upload

Figure 7.1.6.2: After dataset upload

7.1.7 Dataset Info Page

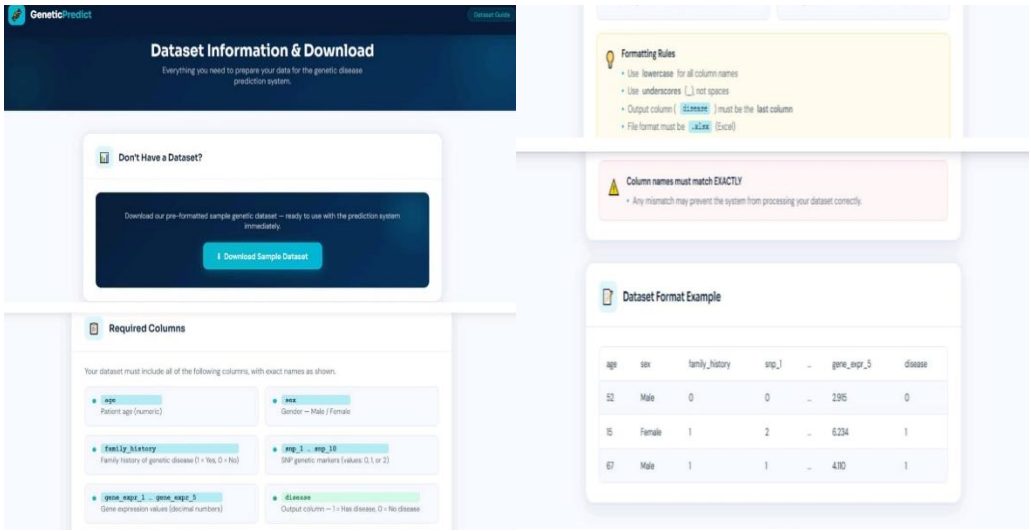


Figure 7.1.7: Dataset Information Page

7.1.8 Prediction Input Fields

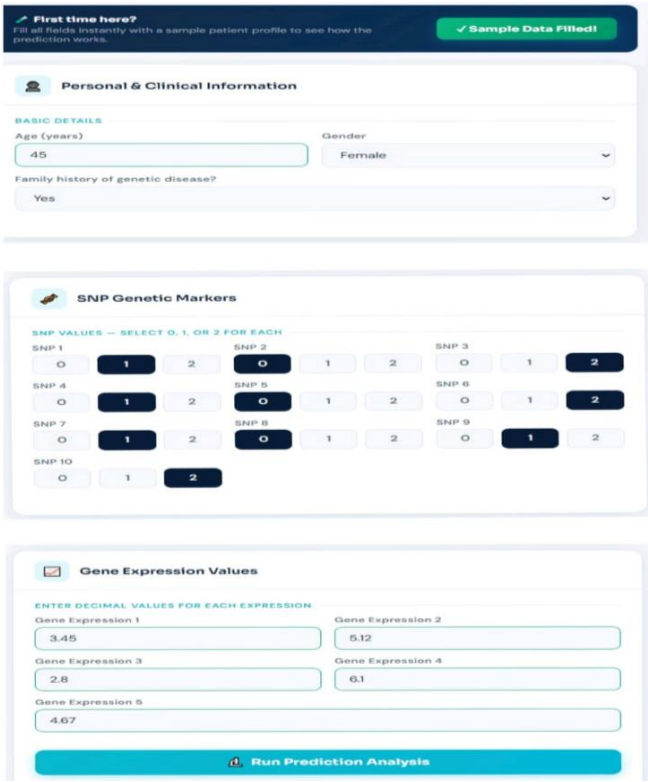


Figure 7.1.8: Input fields

7.1.9 Prediction Output

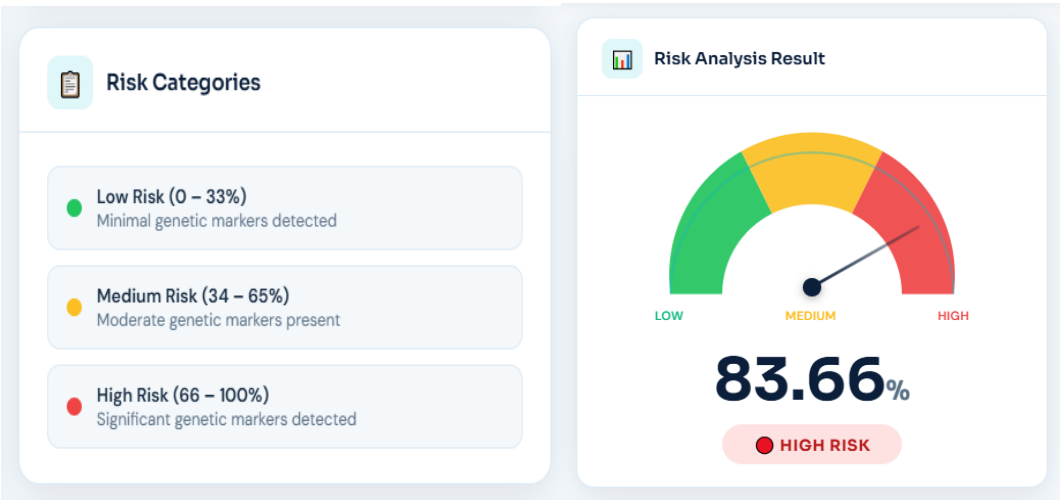


Figure 7.1.9: Final Output

7.1.10 Prediction History page

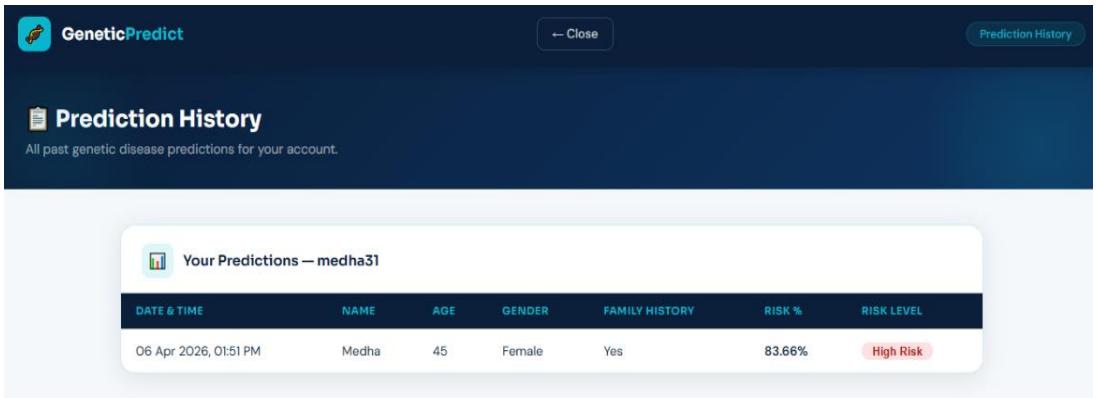


Figure 7.1.10: History page

7.1.11 Navigation Links

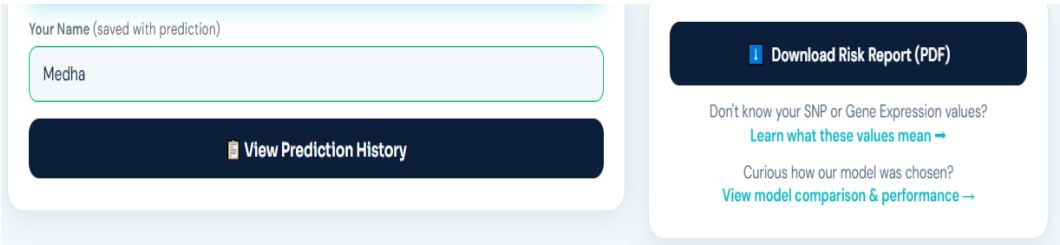


Figure 7.1.11: Navigation

7.1.12 Genetic Information Page

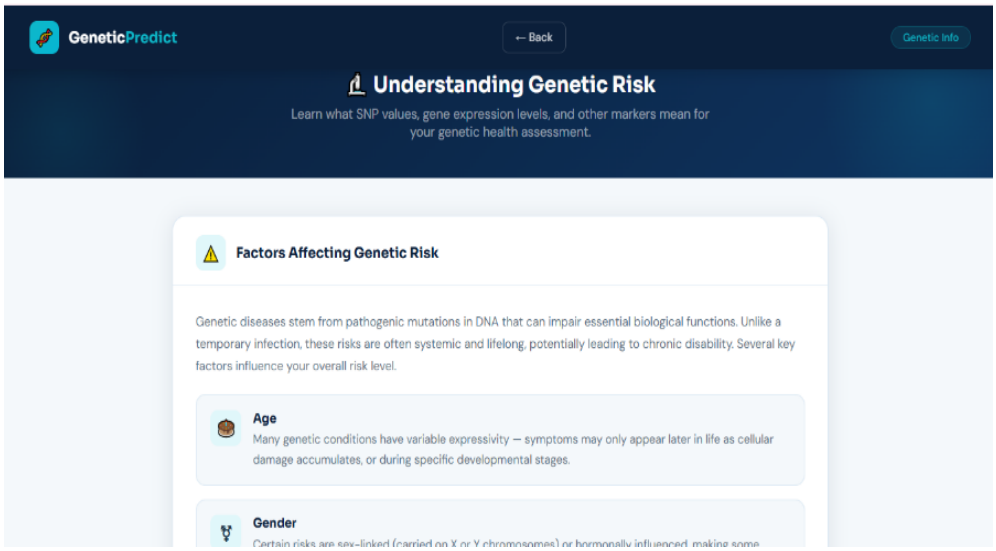


Figure 7.1.12.1: Genetic Information Page

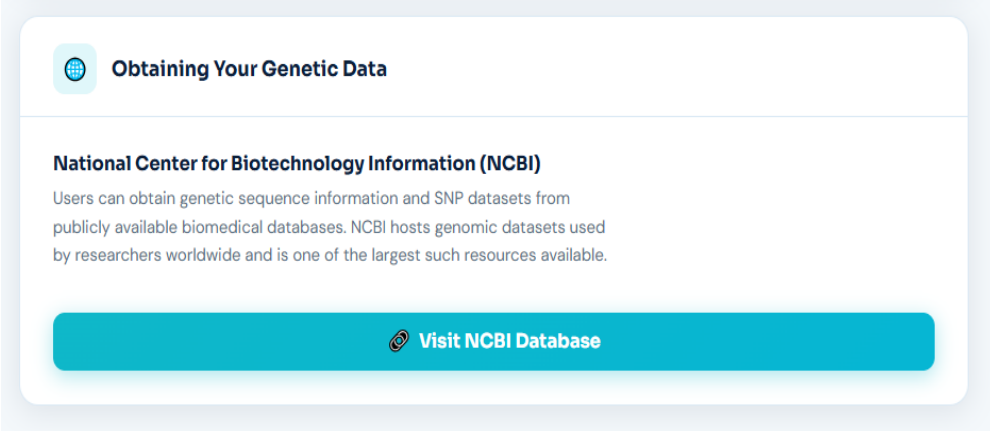


Figure 7.1.12.2: External Integration

7.1.13 Model Comparison Page

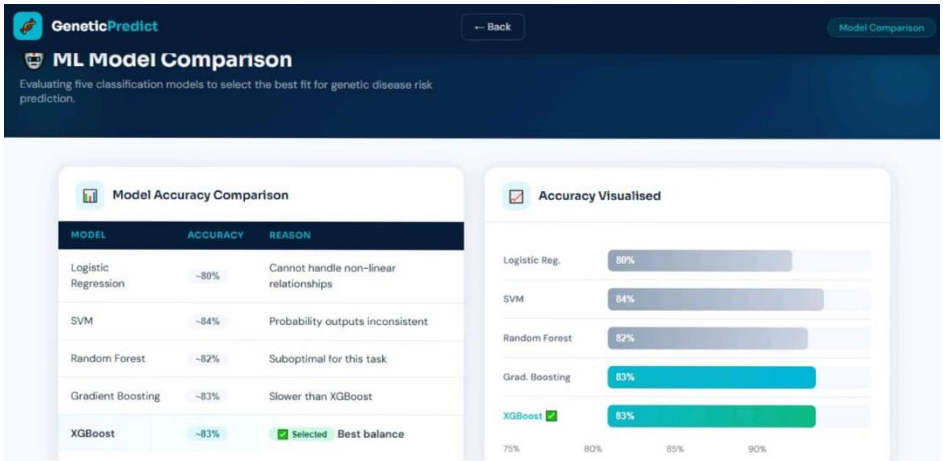


Figure 7.1.13.1: Model Comparison

Figure 7.1.13.2: Classification Report

7.1.14 Backend Output

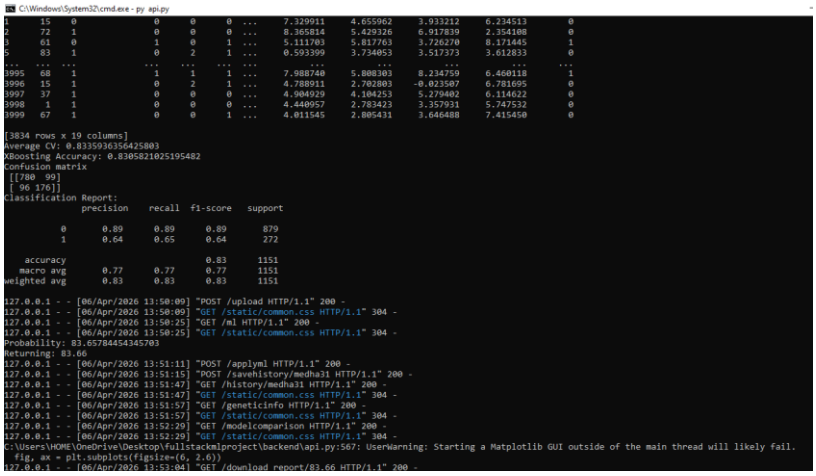
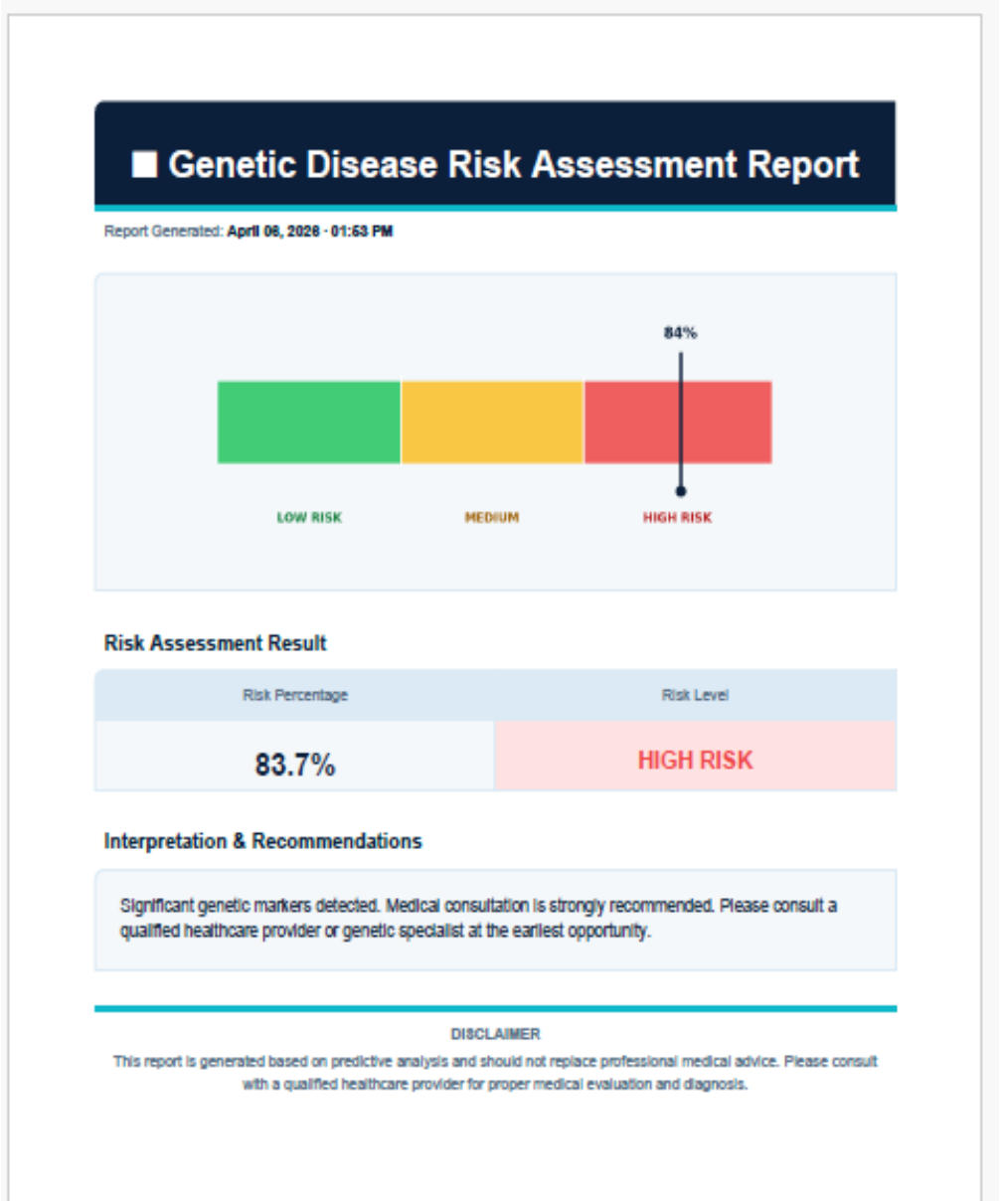


Figure 7.1.14: Backend Output

7.15 PDF Report



Genetic Disease Risk Assessment Report

Report Generated: **April 08, 2026 - 01:53 PM**

Risk Assessment Result

Risk Percentage	Risk Level
83.7%	HIGH RISK

Interpretation & Recommendations

Significant genetic markers detected. Medical consultation is strongly recommended. Please consult a qualified healthcare provider or genetic specialist at the earliest opportunity.

DISCLAIMER

This report is generated based on predictive analysis and should not replace professional medical advice. Please consult with a qualified healthcare provider for proper medical evaluation and diagnosis.

Figure 7.1.15: PDF Report

7.2 Model Performance Evaluation

The performance of the machine learning model was evaluated using various metrics to ensure accuracy and reliability of predictions. The XGBoost algorithm was selected as the final model based on its superior performance compared to other models.

7.2.1 Model Accuracy and Validation

- Cross-Validation Score (Average CV): **0.83**
- Model Accuracy: **0.83**

These results indicate that the model performs consistently well on both training and validation datasets.

1. **Confusion Matrix**

Table 7.2.1: Confusion Matrix

	Predicted: No Risk (0)	**Predicted: Risk (1)**
Actual: No Risk (0)	780	99
Actual: Risk (1)	96	176

2. **Classification Report**

Table 7.2.2: Classification Report

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0 (No Risk)	0.89	0.89	0.89	879
1 (Risk)	0.64	0.65	0.64	272
Metric	**Precision**	**Recall**	**F1-Score**	**Support**
Accuracy	-	-	0.83	1151
Macro Avg	0.77	0.77	0.77	1151
Weighted Avg	0.83	0.83	0.83	1151

7.2.2 Discussion

The model demonstrates strong performance in predicting the majority class (No Risk), with high precision and recall values. However, slightly lower performance is observed for the minority class (Risk), which is a common challenge in imbalanced datasets.

Overall, the model achieves a balanced accuracy of 83%, indicating reliable prediction capability. The results suggest that the system is effective for initial risk assessment, while further improvements can be made by enhancing dataset balance and feature representation.

RESUME

MEDHA U GOWDA

CONTACT

gmedhau@gmail.com· 9141208939· K G Nagar, Bangalore

https://github.com/Medha-U-Gowda

EDUCATION

Diploma in Computer Science & Engineering
Sri Jayachamarajendra Govt. Polytechnic, Bangalore·
Current CGPA: **9.98**
Expected Graduation: 2026

SSLC
Sri Saraswathi Vidya Mandira High School, Bangalore
98.24% 2023

SOFT SKILLS·

Logical Reasoning· Communication

LANGUAGES

· English · Kannada
· Hindi · French

HOBBIES

· Reading novels
· Sketching
· Writing poems
· Researching

CAREER OBJECTIVE

To secure an AI/ML-focused internship where I can apply my Python and machine learning skills, gain hands-on industry experience, strengthen my technical and problem-solving abilities, and contribute meaningfully to impactful projects.

TECHNICAL SKILLS

· Python · Java · HTML
· OOP · DBMS · Data Structures
· Jupyter · VS Code · Git
· Pandas · NumPy · Scikit-learn
· NLTK · spaCy · TensorFlow

PROJECT EXPERIENCE

Early Parkinson's Detection using Voice

· Built an ML model to detect Parkinson's disease via voice recordings — 92% accuracy
· Used Pandas, NumPy & Scikit-learn for preprocessing and model evaluation
· Applied ML techniques to real-world medical data

Automated Assessment Generator

· AI tool generating MCQs & fill-in-the-blanks from text using NLTK and spaCy
· Implemented difficulty ranking and automated distractor generation

TRAINING

· **Samsung Innovation Campus:** Online training in AI & Machine Learning
· **DELL:** Offline training in Excel and AI/ML

CERTIFICATIONS

Infosys Springboard: Git Essentials · ML Fundamentals · Python for Data Science · Data Transformation & Preprocessing · NLP · TensorFlow Introduction

Samsung Innovation Campus Certificate

Simplilearn | SkillUP: Generative AI Studio · Power BI for Beginners · Python for Data Analysis

PHOTO GALLERY

Snapshot of students working in the company and delivering their presentations along with showcasing of their completion certificates.

Figure 9.1: Photo Gallery

REFERENCES

1. **Python Official Documentation**. Available at: https://www.python.org

   Used as the primary programming language for backend and frontend development along with machine learning implementation.

2. **Flask Documentation**. Available at: https://flask.palletsprojects.com

   Used to develop the backend framework and create APIs for communication between frontend and backend modules.

3. **Firebase Documentation.** Available at: https://firebase.google.com

   Used for user authentication and real-time database management.

4. **Scikit-learn Documentation**. Available at: https://scikit-learn.org

   Used for implementing and evaluating initial machine learning models and preprocessing techniques.

5. **XGBoost Documentation**. Available at: https://xgboost.readthedocs.io

   Used as the final machine learning model due to its high performance and ability to handle complex data relationships.

6. **Pandas Documentation**. Available at: https://pandas.pydata.org

   Used for data cleaning, transformation, and handling structured datasets.

7. **Postman Learning Resources**. Available at: https://www.postman.com

   Used for testing API endpoints and validating request-response cycles between frontend and backend.

8. **GeeksforGeeks. Online Tutorials**. Available at: https://www.geeksforgeeks.org

   Used as a supplementary learning resource for general programming concepts and UI enhancement.

APPENDICES

Appendix A: Dataset Used

A structured dataset was used for training and evaluating the machine learning model. The dataset was provided by the organization and was originally sourced from Kaggle (exact source link unavailable).

Dataset Summary:

- **Total samples:** 4,000 rows (Sample preview of first 4 rows shown below)
- **Total features:** 20 columns
- **Target variable:** `disease` (0 = No Risk, 1 = Risk)
- **Format:** Excel (.xlsx)

Column descriptions:

- `patient_id`: Unique identifier for each patient
- `age`: Patient's age in years
- `sex`: Male or Female
- `family_history`: 0 = No, 1 = Yes
- `snp_1` to `snp_10`: Single Nucleotide Polymorphism genetic markers
- `gene_exp_r_1` to `gene_exp_r_5`: Gene expression levels
- `disease`: Target variable (0 = No genetic disease risk, 1 = Genetic disease risk)

Table 11.1: Sample Dataset

patient_id	age	Sex	family_ history		snp_1	…	gene_expr_1	…	disease
1	52	Male	0		0	…	1.669893532	…	0
2	15	Female	0		0	…	4.161756502	…	0
3	72	Male	0		0	…	6.845786018	…	0
4	61	Female	1		0	…	3.74894724	…	1

Note: Due to report length constraints, only the first 4 sample rows are displayed above. The full dataset contains 4,000 rows.

GITHUB LINK OF THE ENTIRE PROJECT

LINK

https://github.com/Medha-U-Gowda/fullstackmlproject